

# Taehyeon Kim

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## RESEARCH PROFILE

**Senior Member of Technical Staff, Holiday Robotics.** My research focuses on efficient LLM inference and adaptive reasoning: how models allocate computation, generate answers, and learn when to reason. I am extending this work to embodied agents that use robot skills and improve through execution feedback.

## SELECTED RESEARCH

- **Contrastive Thinking Decoding [C16] – NeurIPS 2026, accepted; first author.** Steering answer generation in reasoning models.
- **RPO [C15] – ACL 2026, Oral; first author.** Adaptive thinking through root-token policy optimization.
- **Instructive Decoding [C7] – ICLR 2024, Spotlight; co-first author.** Instruction-guided decoding for noisy instructions.
- **Blockwise parallel decoding [C10] – NeurIPS 2024; first author.** Google Research collaboration on draft refinement for faster parallel decoding.

## EDUCATION

### Ph.D., Graduate School of AI

Korea Advanced Institute of Science and Technology (KAIST) - Advisor: Prof. Se-Young Yun  
• Thesis: *Efficient and Effective Inference-time Decoding Strategies for Foundational Models*

Mar 2020 - Feb 2025

### M.S., Graduate School of Data Science

Korea Advanced Institute of Science and Technology (KAIST) - Advisor: Prof. Se-Young Yun  
• Thesis: *Orthogonal Feature Regularization: A Novel Approach for Training Robust Models*

Mar 2018 - Feb 2020

### B.S., Mathematical Sciences & Minor, Intellectual Property Minor Program

Korea Advanced Institute of Science and Technology (KAIST)

Mar 2013 - Feb 2018

## WORK EXPERIENCES

### Holiday Robotics, Embodied AI Team @ Seoul, South Korea

Jun 2026 – Present

- **Senior Member of Technical Staff:** Founding member, Embodied AI Team. Working on robot-use agents, evaluation, and training data for robot learning.

### LG AI Research, Superintelligence Lab @ Seoul, South Korea

Apr 2025 – Jun 2026

- **Tech Lead (Jan–Jun 2026):** Expert-domain AI search system – end-to-end pipeline from retrieval to agentic reasoning.
- **Agentic Orchestration:** Built multi-tool harness with parallel retrieval, LLM query decomposer/planner, and structured examination modeling (novelty & inventiveness judgment), wired via tool-use endpoints.
- **LLM alignment & RL:** SFT, DPO, GRPO-based training (RPO – adaptive thinking via root-token policy, ACL 2026 [C15]); novel DPO for small-large model collaboration; SIMPER for math/coding (in preparation).
- **Research:** Contrastive Thinking Decoding: steering answer generation in reasoning models (NeurIPS 2026, accepted [C16]); adaptive self-taught data generation for RFT, co-corresponding author (NeurIPS 2025 [C12]); LLM negotiation-agent benchmark MERIT (ACL 2026 [C13]); multi-drafter speculative decoding (ACL 2026 Findings [C14]).
- **Retrieval & Reranking:** Trained domain dense retrievers (text-embedding) with cross-encoder rerankers; contrastive learning (hard-negative mining, curriculum sampling).
- **Multi-modal:** Designed VLM fine-tuning experiments for diagram understanding; evaluated cross-modal fusion (image+text) strategies for retrieval grounding.

### Google Research @ NYC, USA

Oct 2023 – Dec 2023

- **PhD Intern:** Fast parallel decoding techniques on large language models (published at NeurIPS 2024 [C10])
- *Working with: Adrian Benton, Ananda Theertha Suresh, and Kishore Papineni*

### Dynamo AI (YC-backed) @ SF, USA

Jan 2023 – May 2023

- **PhD Research Intern:** Privacy-preserving federated learning for enterprise AI; contributed to closing B2B contracts with automotive clients while simultaneously publishing a first-author NeurIPS 2023 paper [C5] (*with Eric Lin*).

### National Institute of Meteorological Sciences (NIMS) @ South Korea

Mar 2022 – Jun 2023

- **Research Manager:** Transformer-based precipitation forecasting; NeurIPS 2022 Competition award (40+ researchers).

### Qualcomm @ Seoul, South Korea

Jun 2021 – Dec 2021

- **PhD Intern:** CV & ML research for autonomous driving (*Working with: Heesoo Myeong*)
- Designing resource-efficient and accurate backbone for ADAS (published at ICML22W [W2], U.S. patent [P1]).

- C16 **Kim, T.**, Jang, Y., Lee, H., Kim, Y. J. & Lee, M. (2026). Contrastive Thinking Decoding: Steering Answer Generation in Reasoning Models, **NeurIPS 2026**, Accepted.
- C15 **Kim, T.**, Lee, H., Jang, Y. & Lee, M. (2026). Efficiently Learning To Reason or Not to Reason: Root-token Policy Optimization for Adaptive Thinking, The 64th Annual Meeting of the Association for Computational Linguistics (**ACL 2026**) (**Oral**), Jul. 2026.
- C14 **Kim, T.\***, Jung, H.\* & Yun, S.. (2026). Multi-Drafter Speculative Decoding with Alignment Feedback, The 64th Annual Meeting of the Association for Computational Linguistics (**ACL 2026 Findings**), Jul. 2026.
- C13 Oh, J., Aghazada, M., Shin, Y., Yun, S. & **Kim, T.†** (2026). MERIT Feedback Elicits Better Bargaining in LLM Negotiators, The 64th Annual Meeting of the Association for Computational Linguistics (**ACL 2026**), Jul. 2026.
- W9 Kim, Y., Yi, E., Kim, M., Yun, S.†, **Kim, T.†** (2025). Guiding Reasoning in Small Language Models with LLM Assistance, **CoLM 2025** Workshop: Test-time Scaling and Reasoning Models.
- C12 Koh, W., Oh, W., Jang, J., Lee, M., Kim, H., Kim, A., Kim, J., Lee, J., **Kim, T.†**, Yun, S.† (2025). AdaSTaR: Adaptive Data Sampling for Training Self-Taught Reasoners, *Advances in Neural Information Processing Systems 38* (**NeurIPS 2025**), Dec. 2025.
- C11 Koh, W., Yoon, J., Lee, M., Song, Y., Cho, J., Kang, J., **Kim, T.**, Yun, S., Yu, Y., Lee, B. (2025). C<sup>2</sup>: Scalable Auto-Feedback for LLM-based Chart Generation, *The 2025 Annual Conference of the Nations of the Americas Chapter of the ACL* (**NAACL 2025**) (**Oral**), May. 2025.
- J2 **Kim, T.**, Kim, D. & Yun, S.. (2024). FLR: Label-Mixture Regularization for Federated Learning with Noisy Labels, Transactions on Machine Learning Research (**TMLR**).
- W8 **Kim, T.\***, Jung, H.\* & Yun, S.. (2024). A Unified Framework for Speculative Decoding with Multiple Drafters as a Bandit, NeurIPS 2024 Workshop: Efficient Natural Language and Speech Processing (ENLSP-IV).
- C10 **Kim, T.**, Suresh, AT., Papineni, K., Riley, M., Kumar, S. & Benton, A., (2024). Accelerating Blockwise Parallel Language Models with Draft Refinement, *Advances in Neural Information Processing Systems 37* (**NeurIPS 2024**), Dec. 2024.
- C9 Ho, N.\*, Bae, S.\* **Kim, T.**, Jo, H., Kim, Y., Schuster, T., Fisch, A., Thorne, J. and Yun, S. (2024). Block Transformer: Global-to-Local Language Modeling for Fast Inference, *Advances in Neural Information Processing Systems 37* (**NeurIPS 2024**), Dec. 2024.
- C8 Yi, E.\* **Kim, T.\***, Jeung, H., Chang, D. and Yun, S. (2024). Towards Fast Multilingual LLM Inference: Speculative Decoding and Specialized Drafters, *The 2024 Conference on Empirical Methods in Natural Language Processing* (**EMNLP 2024**), Nov. 2024.
- W7 **Kim, T.**, Suresh, AT., Papineni, K., Riley, M., Kumar, S. & Benton, A., (2024). Exploring and Improving Drafts in Blockwise Parallel Decoding, ICML 2024 Workshop: Efficient Systems for Foundation Models (Es-FoMo).
- C7 **Kim, T.\***, Kim, J.\* **Kim, T.\***, Lee, G.\* & Yun, S.. (2024). Instructive Decoding: Instruction-Tuned Large Language Models are Self-Refiner from Noisy Instructions, *Twelfth International Conference on Learning Representations* (**ICLR 2024**) (**Spotlight, Top 5%**), May. 2024.
- C6 Yang, Y., **Kim, T.** & Yun, S.. (2024). Leveraging Normalization Layer in Adapters with Progressive Learning and Adaptive Distillation for Cross-Domain Few-Shot Learning, *The 38th Annual AAAI Conference on Artificial Intelligence* (**AAAI 2024**), Feb. 2024.
- C5 **Kim, T.**, Lin, E., Lee, J., Lau, C. & Mugunthan, V.. (2023). Navigating Data Heterogeneity in Federated Learning: A Semi-Supervised Federated Object Detection, *Advances in Neural Information Processing Systems 36* (**NeurIPS 2023**), Dec. 2023.
- C4 Gruca et al. Weather4cast at NeurIPS 2022: Super-Resolution Rain Movie Prediction under Spatio-temporal Shifts, *Proceedings of the NeurIPS 2022 Competition and Demonstration Track, in Proceedings of Machine Learning Research* 220:292-313 (2022).
- W6 **Kim, T.**, Kang, S., Shin, H., Yoon, D., Eom, S., Shin, K. & Yun, S.. (2022). Region-Conditioned Orthogonal 3D U-Net for Weather4Cast Competition, NeurIPS 2022 Workshop: Weather4Cast Competition.
- W5 Eom, S., **Kim, T.** & Yun, S.. (2022). Layover Intermediate Layer for Multi-Label Classification in Efficient Transfer Learning, NeurIPS 2022 Workshop: Has it Trained Yet? (HITY).
- W4 Shin, J., **Kim, T.** & Yun, S.. (2022). Revisiting the Activation Function for Federated Image Classification, NeurIPS 2022 Workshop Federated Learning.
- W3 **Kim, T.** & Yun, S.. (2022). Supernet Training for Federated Image Classification under System Heterogeneity, ICML 2022 Workshop on Dynamic Neural Networks (**Oral**).
- J1 **Kim, T.** & Yun, S.. (2022). Revisiting Orthogonality Regularization: A Study for Convolutional Neural Networks in Image Classification, **IEEE Access**, Jun. 2022.
- W2 **Kim, T.**, Myeong, H. & Yun, S.. (2022). Revisiting Architecture-aware Knowledge Distillation: Smaller Models and Faster Search, ICML 2022 Hardware Aware Efficient Training (HAET) Workshop, July 2022.
- C3 **Kim, T.\***, Ko, J.\* **Kim, T.\***, Cho, S., Choi, J. & Yun, S.. (2021). FINE Samples for Learning with Noisy Labels., *Advances in Neural Information Processing Systems 34* (**NeurIPS 2021**), Dec. 2021.
- C2 **Kim, T.\***, Oh, J.\* **Kim, T.\***, Kim, N., Cho, S., & Yun, S. Y. (2021). Comparing Kullback-Leibler Divergence and Mean Squared Error Loss in Knowledge Distillation. *In the 30th International Joint Conference on Artificial Intelligence* (**IJCAI 2021**), Aug. 2021 (acceptance rate: 13.9%)

- W1 **Kim, T.\***, Ahn, J.\*, Kim, N.\*, & Yun, S. (2020). Adaptive Local Bayesian Optimization Over Multiple Discrete Variables. Workshop at NeurIPS 2020 Competition Track on Black-Box Optimization Challenge, Dec. 2020.
- C1 **Kim, T.**, Kim, J. & Yun, S. (2019). Efficient Model for Image Classification With Regularization Tricks. *Proceedings of the NeurIPS 2019 Competition and Demonstration Track, in Proceedings of Machine Learning Research 123:13-26*.
- T **Kim, T.**. Orthogonal feature regularization : a novel approach for training robust models, Korea Advanced Institute of Science and Technology (KAIST)

## ONGOING MANUSCRIPTS

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- U3 **Kim, T.** et al. (2025). Masked-SIMPER: Improving Collaborative Reasoning, In preparation.
- U1 Bartholet, M.\*, **Kim, T.\***, Beuret, A. Yun, S. & Buhmann, J. (2024). Hypernetwork-Driven Model Fusion for Federated Domain Generalization, Under review.

## US PATENTS

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- P1 **Kim, T.**, Myeong, H. (2023). Trust-Region Aware Architecture Distillation for Sample-Efficient Neural Architecture Search, Qualcomm Inc..

## AWARDS @ COMPETITIONS

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- 4th Award in NeurIPS 2022 Competition: Weather4Cast** Dec 2022  
 • Homepage: <https://www.iarai.ac.at/weather4cast/>  
 • Subjects: Precipitation Forecast, Segmentation, Video
- 8th Award in NeurIPS 2020 Black-Box Optimization Challenge** Dec 2020  
 • BBO-challenge homepage: <https://bbochallenge.com/>  
 • Subjects: Auto-ML, Bayesian Learning, Hyperparameter Optimization.
- 2nd & 3rd Awards in NeurIPS 2019 MicroNet Challenge, CIFAR-100 Track.** Dec 2019  
 • MicroNet-challenge homepage: <https://micronet-challenge.github.io/>  
 • Subjects: Image Classification, Model Compression.

## HONORS

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- Technical Reviewer (ICML '26): Gold** Jul 2026  
 Top 25% of Reviewer (Free Registration).
- Google Conference Scholarship** Dec 2024  
 Accelerating Blockwise Parallel Language Models with Draft Refinement
- Google Cloud Platform (GCP) Credit Award** Mar 2024  
 Gemma Academic Program
- Winner, 2022 Qualcomm AI Fellowship** Oct 2022  
 FINE Samples for Learning with Noisy Labels.
- Best Poster Awards, KAIST AI 21/22 Workshop** Jan 2022  
 1. FINE Samples for Learning with Noisy Labels.  
 2. Comparing Kullback-Leibler Divergence and Mean Squared Error Loss in Knowledge Distillation

## LEADERSHIP

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- Tech Lead & Squad Lead @ LG AI Research** Sep 2025 – Jun 2026  
 • Led a team of 10 researchers (5 PhDs) across retrieval, LLM alignment, and agentic systems.  
 • Defined the research roadmap, mentored junior researchers, and managed cross-team project delivery.
- 1st Representative of doctoral students @ KAIST AI** Mar 2021 – Feb 2022  
 • Construct organizations  
 • Being an intermediary between the professors and the students
- Lab master @ OSI LAB** Mar 2020 – Feb 2021  
 • Optimization and Statistical Inference Laboratory (OSI LAB), KAIST.  
 • Construct organizations (e.g., seminar, lab policy, funding)  
 • Being an intermediary between the advisor and the students
- Vice Captain @ KAIST Representative Cheerleading Group** Oct 2014 – Nov 2015  
 • A.K.A. Encouraging Leaders of KAIST (ELKA)  
 • KAIST festival planning and promotion  
 • 2015 Gwangju Summer Universiade - University Student U Cheering Festival **1st prize**  
 • Team building, Funding, Management.

## INVITED TALKS

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- Naver AI Lab Seminar** Jan 2025  
 • LLM Inference-Time Strategies and its Beyond.

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|-----------------------------------------------------------------------------------------------------|----------|
| <b>Microsoft Research Seminar</b>                                                                   | Jan 2025 |
| • Instructive Decoding and its Beyond.                                                              |          |
| <b>Squeezebits Efficient AI Seminar</b>                                                             | Jun 2024 |
| • Towards Fast LLM Inference: Speculative Decoding and Efficient Architectures                      |          |
| <b>KCC 2024</b>                                                                                     | Jun 2024 |
| • Navigating Data Heterogeneity in Federated Learning: A Semi-Supervised Federated Object Detection |          |
| <b>HyperConnect Seminar</b>                                                                         | Jan 2023 |
| • Personalized Federated Learning on System Heterogeneity under Label Noise                         |          |
| <b>KSC 2022</b>                                                                                     | Dec 2022 |
| • FINE Samples for Learning with Noisy Labels                                                       |          |
| <b>Naver Clova AI Seminar</b>                                                                       | Jun 2019 |
| Orthogonal Feature Regularization: A Novel Approach for training robust model                       |          |

## SKILLS & OTHERS

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|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reviewer</b>  | <b>Conference:</b> NeurIPS {2022, 2023, 2024}, ICML {2022, 2023, 2024}, AutoML-Conf 2022, ECCV 2022, KDD 2023, ICLR {2024, 2025}, ARR 2024.<br><b>Journal:</b> Pattern Recognition. |
| <b>Coding</b>    | Python, PyTorch, $\LaTeX$ , ...                                                                                                                                                     |
| <b>AI in Art</b> | Surplus human, Vania Oh, Junbeom Shin, <b>Taehyeon Kim</b> , 2022.<br>• AI with Weird Wonderland, 22.03.10 - 22.03.16, CELINE PARK GALLERY, Organized by Next interface lab.        |

## TEACHING EXPERIENCES

|                                                                                                     |                     |
|-----------------------------------------------------------------------------------------------------|---------------------|
| <b>AI Lecturer @ ELICE, Mathematics in ML</b>                                                       | Jan 2022 – Feb 2022 |
| • Teaching the mathematics for gradient descent in ML (4 lectures, theory and code quiz, exams)     |                     |
| • Attendee: LG Employees.                                                                           |                     |
| <b>AI Mentor @ Curinc, Introduction to Deep Learning</b>                                            | Jun 2021 – Jul 2021 |
| • Teaching the overview of deep learning in Curinc, Seoul, Korea (70 hours during 7 weeks)          |                     |
| • Attendee: Undergraduate at UC Berkeley, Boston University, and Florida International University.  |                     |
| <b>TA @ LG, Computer Vision &amp; Deep Learning</b>                                                 | Oct 2020 – Nov 2020 |
| • Attendee: LG Employees in the LG AI CAMP Module(3), LG, Academy, Korea                            |                     |
| • Image classification and semantic segmentation using public COVID dataset in Kaggle.              |                     |
| <b>Advisory Committee @ National Science Museum, Deep Learning</b>                                  | Mar 2020 – Oct 2020 |
| • AI Exploration Program, National Science Museum, Korea.                                           |                     |
| • Attendee: Advanced students (Science High School)                                                 |                     |
| • Subjects: reinforcement learning, object detection, image classification, evolutionary algorithm. |                     |
| <b>TA @ KAIST Data Science, Introduction to Deep Learning</b>                                       | Mar 2019 – Jul 2019 |
| • Dept. Knowledge Service Engineering, KAIST, Korea.                                                |                     |
| • Attendee: Graduate Students                                                                       |                     |
| <b>TA @ LG, Computer Vision &amp; Deep Learning</b>                                                 | Jan 2019 – Jan 2019 |
| • Theories and practices for deep learning in the LG AIB Intermediate CAMP, LG Academy, Korea.      |                     |
| <b>Lecturer @ Samsung, Python Basics</b>                                                            | Jan 2019 – Jan 2019 |
| • Python code implementation in the Samsung SW Academy Start CAMP, Samsung SW Academy, Korea.       |                     |
| • Subjects: Chatbot, basic python (e.g., for loop, condition)                                       |                     |
| <b>Lecturer @ KAIST, Regular Leadership Course</b>                                                  | Mar 2017 – Dec 2017 |
| • KAIST Lecture: Personality/Leadership 3 - Liberal Arts Required (2017 Spring & 2017 Fall)         |                     |
| • Subjects: Sports leadership, Team leadership.                                                     |                     |
| • Students: undergraduate freshman in KAIST                                                         |                     |

## Scholarship & Fellowship

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|--------------------------------------------------------------------------------|---------------------|
| <b>Graduate School of AI, KAIST Funding</b>                                    | Mar 2020 – Present  |
| \$ 20,000 per year                                                             |                     |
| <b>National scholarship for graduate studies, Korea Student Aid Foundation</b> | Mar 2018 – Feb 2020 |
| \$ 16,000 per year                                                             |                     |
| <b>Full Academic scholarship, KAIST</b>                                        | Mar 2013 – Aug 2017 |
| \$ 3,000 per year                                                              |                     |